

 TA24 Engineering Work Order for In-kind Replacement (Non MOC) Инженерная Инструкция для Работ КР24 по Аналогичной Замене (Без КЗИ)									
Anomaly Report: Отчет об аномалии:	N/A	EWR #: # ЗИР:	22-0294	Subsystem: Подсистема:	TL	Line №: № Линии:	051-TL01-1/2-150H03		
	Area: Участок:	KTL1	WO: РЗ:	1623137	P&ID #: СТИКИП	1-500_1-32A	Title: Название:	TLR #21069 elimination/ Устранение TLR #21069	
TENGIZCHEVROIL ТЕНГИЗШЕВРОЙЛ	Unit: Установка:	U500.1	Drawing/Sketch No: Чертеж/Схема No:		L-SK-001				
Standard of Acceptance: Стандарт Приемки:	B31.3		TCO Specification: Спецификации ТШО:		W-ST-2025		TCO Approved WPS: Утвержденная Сварочная ПроцедураТШО:		20 AW
Is wall thickness measurement required at tie-in points prior to start works (to verify existing pipe sound wall thickness) Требуется ли замер толщины стенки на точках врезки перед началом работ (для определения приемлемой толщины существующего трубопровода)			YES		Is 100% PT/MT required for weld bevel ends prior to welding Требуется ли 100% ККД/МПД для проверки кромок перед сваркой			YES/ДА	
Shop Hydro-test Испытание Гидро-тестом в Цеху		NO/HET barg	Piping NDT НК Трубопровода		RT 5% VT/BK 100%	Coating System #: № Системы Покрытия:	12.1	Hydrogen Bake out Водородный Отжиг	NO/HET
Field Hydro-test Испытание Гидро-тестом на Участке		Service Test barg	Piping Support Structural NDT НК Опорной Конструкции Трубопровода		NA/НП	Insulation type and thickness: Тип и толщина изоляции:	HC30	PMI for Alloy materials (SP 20) ХАМ для Материалов из сплавов (ПТБ 20)	NA/НП
HARDNESS TESTING (Maximum up to 200HB) ОПРЕДЕЛЕНИЕ ТВЕРДОСТИ (Максимум до 200HB)		NO/HET	Stress Relief (PWHT) Снятие Напряжений (ПСТО)		NO/HET	Ferrite Test Определение на содержание Феррита			NO/HET
Notes Примечания									

Job Description
Описание работ

See next pages for written steps.

Смотрите следующие страницы для расписанных шагов.

Approved by Потверждено	Name Имя	Signature Подпись
Originated by: Подготовлено:	DE Engineer: Инженер ИПГ:	Ayan Bezbalinov Digitally signed by Ayan Bezbalinov Date: 2023.02.21 07:20:38 +05'00'
Approved by: Утверждено:	DE Mentor: Консультант ИПГ:	Nurzhan Zhakishev Digitally signed by Nurzhan Zhakishev Date: 2023.02.22 14:52:08 +05'00'
Reviewed by: Рассмотрено:	RE QM Lead: Вед. Специалист КК Отдела Надежности:	
Reviewed by: Рассмотрено:	FER Authorized Inspector: Уполномоченный инспектор НСО:	
Reviewed by: Рассмотрено:	TTT Representative: Представитель группы КР:	
Reviewed by: Рассмотрено:	TA Operations Representative: Представитель отдела КР Эксплуатации:	
Reviewed by: Рассмотрено:	DE TA Supervisor: Начальник группы КР ИПГ:	
Distribution List Список Рассылки		

desupta; reqaqcta1; reqaqc4; reinfo; tadoccon; tadccsgp;

1. INTRODUCTION

1.1. PRESENT SITUATION:

Leak on low pressure condensate line was fixed by TLR. Repair on-the-run is impossible, this scope transferred to TA24 scope of work.

1.2. PROPOSED SOLUTION:

Dismantle clamp and replace damaged pipe section.

2. SCOPE OF WORK

- CONTRACTOR shall strictly follow all relevant TCO Safety Instructions in the progress of works, all of which are readily obtainable from the TCO Intranet.
- CONTRACTOR shall follow all Quality Management (QM) requirements designated in the Quality Control package.

2.1. PRELIMINARY SITE WORK:

- 2.1.1. Prior to the start of the mechanical activities, perform wall thickness testing at TP-001 and TP-002 to confirm that wall thickness is adequate for welding. Results shall be reviewed and approved by the Fixed Equipment Reliability (FER) Unit Inspector.

NOTE: Perform the wall thickness test by marking out and recording a matrix as directed by the TCO FER Unit Inspector.

2.2. SITE WORK:

- 2.2.1. Perform relevant job safety assessments (PPHA & JSA) and obtain required Permits-to-Work (PTWs).
- 2.2.2. Demarcate work area as necessary and erect scaffolding for access.

NOTE: **TCO Operations** shall isolate, depressurize, drain and ready for Hot Work, Equipment and Piping associated with this Project, if and as applicable, and shall hand it over to CONTRACTOR; fully accessible, safe and Locked-Out-Tagged-Out (LOTO) as scheduled and agreed between TCO Operations and CONTRACTOR.

- 2.2.3. Remove all cladding and insulation to the extent required.
- 2.2.4. Cold Cut and remove existing piping shown on attached sketch L-SK-001.
- 2.2.5. Bevel cut ends of existing piping at the tie-in points TP-001 and TP-002 for field welding as shown on attached sketch L-SK-001.
- 2.2.6. Inspect the prepared weld bevel ends 100% VT and 100 % MT/PT for defects.

NOTE: Perform inspection of weld bevels and nipo-flange landings PRIOR to welding.

- 2.2.7. Install piping as specified on attached sketch L-SK-001. For field welds use TCO-approved WPSs indicated on attached sketch L-SK-001.
- 2.2.8. Perform NDE of field welds, as specified on attached sketch L-SK-001.
- 2.2.9. Perform surface preparation and touch up field welds and any other coating damages in accordance with TES COM-SU-4743-TCO and coating systems per TES COM-SU-5191-TCO.
- 2.2.10. All QM piping passport documentation (A-category Checklists) to be verified and signed by TCO-approved QM Inspector prior to PSSR, but not later than 24 hours after mechanical scope completion.
- 2.2.11. Advise DE T/A Engineer at ext. 3972 about work completion and invite all necessary people to PSSR.
- 2.2.12. Job Pack Coordinator shall submit the completed PIC check-list to Operations department after being signed by Quality Management group. (Form is provided by QM Dept.).
- 2.2.13. Conduct PSSR and close out A-category punch points.
- 2.2.14. Following PSSR reinstate / install insulation and cladding, as per TES IRM-SU-1381-TCO and as specified on attached sketch L-SK-001.
- 2.2.15. Close out all B-category punch points and remove associated scaffolding.
- 2.2.16. Clean up the work area and demobilize from site.

2.3. AFTER COMPLETION OF WORK

- 2.3.1. Perform service test.
- 2.3.2. Final as-built package (including B-category Checklist) shall be submitted to TCO QM group within 2 weeks after completion of the whole scope of works.
- 2.3.3. Return excess material to TCO Warehouse.

3. REFERENCED TCO SPECIFICATIONS:

A-ST-2008	Basic Engineering Data
COM-SU-4743-TCO	External Coatings
W-ST-2025	Welding, PWHT & NDE of Piping
IRM-SU-1381-TCO	Thermal Insulation for Hot Lines, Vessels and Exchangers
COM-SU-5191-TCO	Coating Systems